A Minimalistic Approach to Real-Time Audio Steganography Using Pseudo-Random Sample Embedding

Radhanjan Neelamraju
23BCI0126
Department of Information Security, SCOPE
VIT Vellore
Vellore, India
Radhanjan.neelamraju2023@vitstudent.ac.in

Rhutvik Hegde
23BCI0079
Department of Information Security, SCOPE
VIT Vellore
Vellore, India
Rhutvik.hegde2023@vitstudent.ac.in

1. Abstract

Traditional digital steganography, while effective at concealing data within media, typically operates on pre-recorded files, rendering it unsuitable for modern, real-time communication demands. This paper addresses the critical need for undetectable, live message exchange by proposing and evaluating a minimal Real-Time Dynamic Audio Steganography Framework.

The core innovation lies in establishing a highly synchronized, low-latency covert channel within a live audio stream transmitted over a TCP/IP network, surpassing the limitations of static, offline methods. The system utilizes a lightweight, modular architecture consisting of a Transmitter and a Receiver. The Transmitter captures live 44.1 kHz PCM audio, segments it into small 1024-sample frames, and employs a Pseudo-Random Number Generator (PRNG) with a shared key to deterministically select sample indices for Least Significant Bit (LSB) embedding.

This synchronized PRNG indexing ensures perfect alignment between sender and receiver without explicit signaling, maximizing stealth and minimizing delay. The framework's design also targets future integration of AES encryption for confidentiality and psychoacoustic adaptive embedding to further optimize capacity and imperceptibility based on local frame energy.

Experimental results validate the feasibility and superior performance of the prototype. The framework achieved an exceptionally low average end-to-end latency of approximately 10.87 milliseconds (ms), confirming its operational capability in near real-time conditions. The maximum theoretical embedding capacity was established at 1378.12 bits per second (bps), sufficient for continuous text transmission.

Crucially, the stego-audio quality was excellent, yielding an average Signal-to-Noise Ratio (SNR) of 103.33 dB, signifying negligible acoustic distortion. Furthermore, the reliance on TCP and perfect synchronization resulted in a Bit Error Rate (BER) of 0.000%, demonstrating complete message integrity and channel reliability.

In conclusion, this project successfully demonstrates a practical, high-fidelity method for implementing covert communication in live audio streams. The validated low-latency and high-integrity performance paves the way for advanced future work, including the integration of adaptive masking and strong cryptography to enhance the security and robustness of the covert channel.

Keywords: Audio Steganography, Real-Time Communication, Covert Channel, Least Significant Bit (LSB), Pseudo-Random Number Generator (PRNG), Low Latency, Network Steganography, TCP/IP Streaming, High-Fidelity Audio, Signal-to-Noise Ratio (SNR), Bit Error Rate (BER), Digital Signal Processing (DSP), Acoustic Imperceptibility

2. Introduction

In today's digital landscape, communication security extends far beyond conventional cryptography. While encryption effectively secures the *content* of a message, the presence of overtly encrypted traffic often serves as a signal, drawing unwanted attention from analysts and adversaries. Steganography, in contrast, offers a powerful, stealth-focused defense mechanism by concealing the very existence of information itself. This is achieved by embedding secret data imperceptibly within ordinary, innocuous media—such as images, videos, or audio—rendering the presence of a hidden message undetectable to human senses or standard statistical analysis tools [1], [8].

Among the various cover media available, audio provides a particularly advantageous carrier for covert communication. Subtle modifications within the temporal or frequency domains of a sound wave can mask hidden information by exploiting the high redundancy and inherent masking properties of the human auditory system [2]. However, the foundational research in audio steganography has historically focused on offline processing. Traditional methods typically operate on static, prerecorded audio files, embedding the secret data into digital samples before the file is stored or transferred [3], [4]. While these file-based techniques are robust, their inherent latency and requirement for full file processing fundamentally preclude them from supporting the demands of real-time covert communication, which is critical for modern applications requiring instant, continuous, and undetectable message exchange across networks [6].

This project directly addresses the significant performance and application gap left by traditional, static approaches by developing a minimal real-time audio steganography system. This system enables hidden messages to be continuously transmitted live between two systems over a local network, transforming the passive steganographic channel into an active, dynamic covert communication link. The core technical mechanism relies on the Least Significant Bit (LSB) modification technique, chosen for its computational simplicity and low perceptual impact, making it ideal for low-latency operation. Crucially, we employ a pseudo-random number generator (PRNG) seeded with a shared secret key to maintain strict, deterministic synchronization [7]. This PRNG dictates the exact, non-sequential LSB indices in the live-captured audio frames where the secret bits are embedded, ensuring that the receiver can perfectly mirror the extraction process without requiring disruptive signaling or complex handshake protocols. The resulting "stego-audio" stream is transmitted via TCP and decoded by a synchronized receiver in real time, demonstrating the feasibility of live covert communication [6].

The design goals guiding this prototype are tripartite: to maintain imperceptibility, ensure seamless real-time synchronization, and minimize end-to-end latency to below audible thresholds. The choice of a small, 1024-sample audio frame size is a direct optimization toward achieving this low latency, ensuring the entire process—capture, embedding, transmission, reception, and playback—occurs in near real-time. Although this initial prototype focuses on proving the fundamental feasibility of live LSB-PRNG embedding and omits advanced security and error-correction features for simplicity, the successful results lay a solid foundation for more robust implementations. The presented architecture is designed to accommodate seamless future enhancements that will elevate its utility.

Future work planned for the framework will focus on three key enhancements. First, the integration of AES-based encryption is essential to ensure message confidentiality, rendering the concealed data unintelligible even if the covert channel is detected [10]. Second, psychoacoustic adaptive embedding will be incorporated, utilizing auditory masking models to dynamically adjust the LSB modification depth (e.g., from the 1st LSB to the 2nd or 3rd LSB) based on the local energy and complexity of each audio frame, thereby maximizing payload capacity while rigorously preserving imperceptibility [9], [14]. Finally, the inclusion of forward error correction (FEC) codes will be necessary to improve message integrity and robustness, providing resilience against potential network disturbances, noise, or packet loss inherent to non-ideal network environments [22]. In summary, this minimal real-time framework not only successfully bridges the gap between traditional steganography and live networking but also provides a versatile and scalable architecture for secure, dynamic covert communication.

3. Literature Review

Steganography has evolved from ancient physical methods to sophisticated digital techniques embedded in multimedia. Foundational work on Least Significant Bit (LSB) modification established a basis for both image and audio steganography, valued for its simplicity and low perceptual impact [1], [2].

A. Audio Steganography Techniques

Audio steganography commonly manipulates the properties of digital audio signals, exploiting redundancy and psychoacoustic masking [3], [5]. Techniques such as phase coding [4], echo hiding [5], and spread spectrum [6] enhance robustness against attacks and compression. Bender et al. [4] introduced early phase-based methods that improved resistance to signal distortion, while Brassil et al. [5] explored echo-based hiding to achieve perceptual transparency.

B. Real-Time and Network-Based Approaches

Traditional research focused on static files, but emerging studies have explored **r**eal-time and streaming steganography**,** embedding data during live transmission [6], [7]. Gopalan and Kalaiselvi [7] proposed PRNG-based embedding for real-time audio streams, using synchronized key sequences to maintain alignment between sender and receiver without additional signaling. However, most existing implementations suffer from high buffering delay or require complex synchronization, reducing real-time feasibility.

C. Synchronization and PRNG-Based Embedding

Synchronization between transmitter and receiver is a core challenge in real-time steganography. PRNG-based synchronization ensures both sides use identical sample indices for embedding and extraction [7], [8]. This approach eliminates explicit synchronization signals, improving stealth and efficiency. Studies confirm that PRNG-driven LSB embedding offers a favorable balance between capacity, imperceptibility, and computational simplicity [8].

D. Future Enhancements: Encryption and Adaptive Embedding

Recent work integrates encryption and psychoacoustic models to strengthen security and perceptual transparency [9], [10], [14], [15]. AES encryption ensures confidentiality even if the covert channel is detected [10], while adaptive embedding adjusts hiding strength according to auditory masking thresholds [14]. Modern extensions also incorporate deep learning [16]–[18] and generative adversarial networks (GANs) for optimized embedding under perceptual constraints [16], [18], [25].

In summary, although digital audio steganography has matured significantly, real-time implementations remain rare and challenging. This project contributes a lightweight, synchronized, and practical prototype that demonstrates the feasibility of live covert communication using simple yet effective LSB-PRNG embedding.

4. Methodology

4.1 Real-Time Dynamic Audio Steganography Framework

The proposed system implements a dynamic, full-duplex real-time audio steganography framework capable of concealing encrypted data within live audio streams. Unlike static, file-based approaches, this design enables continuous covert transmission during active communication, maintaining imperceptibility and synchronization over TCP/IP networks. The framework integrates four key modules—Audio Capture, Encryption, Dynamic Embedding, and Extraction—each optimized for low-latency operation.

The system architecture (Fig. 1) consists of two endpoints: a Transmitter and a Receiver, connected via a dedicated TCP socket stream. The audio signal from the microphone is processed frame-by-frame; encrypted message bits are embedded dynamically into least significant bits (LSBs) according to frame energy, and transmitted to the receiver, which extracts and decrypts the hidden message in real time.

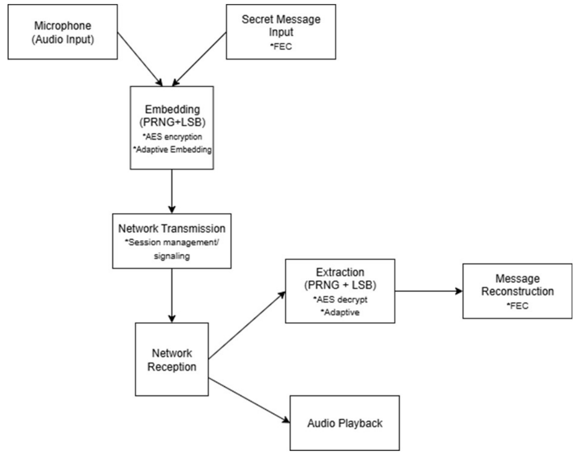

Figure 1. System Architecture of the Proposed Real-Time Audio Steganography Framework

4.2 Transmitter System

The transmitter begins by capturing live audio from the system microphone using the PyAudio interface.

- Sampling rate: 44.1 kHz
- Sample depth: 16-bit PCM
- Frame size: 1024 samples (~23 ms)

These parameters provide a trade-off between latency and fidelity. Each frame is immediately passed to the embedding pipeline without buffering delays.

Before embedding, the plaintext message is encrypted using the Advanced Encryption Standard (AES) in Cipher Block Chaining (CBC) mode [10]. The message text is converted to bytes, padded to 16-byte blocks, and encrypted using a shared symmetric key and Initialization Vector (IV). The ciphertext is serialized into a bitstream suitable for embedding.

Algorithm 1. AES-Based Message Encryption

```
Input: plaintext M, key K, initialization vector IV
Output: encrypted bitstream B_enc
1:  Convert M → byte sequence
2:  Pad to 16-byte blocks (PKCS7)
3:  C ← AES_Encrypt_CBC(M, K, IV)
4:  B_enc ← ConvertBytesToBits(C)
5:  return B_enc
```

This encryption ensures that even if the stego-audio is intercepted, the concealed data remains unintelligible without the secret key.

This is the heart of the framework. The module employs a dynamic adaptive LSB embedding strategy guided by frame energy and variance.

- High-energy regions (loud or complex frames) allow deeper bit modification (2nd–3rd LSB).

- Low-energy regions limit changes to the 1st LSB or skip embedding entirely to prevent audible distortion.

A pseudo-random sequence generator (PRNG) seeded with a pre-shared key determines the exact sample indices used for embedding, ensuring deterministic synchronization between sender and receiver [7].

Algorithm 2. Dynamic LSB Embedding

```
Input: frame F[1..N], bitstream B, PRNG seed S
Output: stego frame F'
E ← Energy(F)
if E > threshold_high then depth ← 3
else if E > threshold_low then depth ← 2
else depth ← 1
indices ← PRNG(S).sample(N, |B|)
for i in 1..|B| do
    bit_pos ← depth(i)
    F'[indices[i]] ← SetLSB(F[indices[i]], B[i], bit_pos)
return F'
```

This adaptive mechanism improves imperceptibility while maintaining sufficient embedding capacity [2], [14].

The Network Transmission Module manages real-time transport of stego-frames via TCP, chosen for its ordered and reliable delivery. Each stego-frame is serialized into bytes and transmitted immediately after embedding, maintaining continuous streaming with negligible buffering delay. TCP's integrity checks ensure synchronization across both endpoints, critical for successful extraction.

4.3 Receiver System

The receiver mirrors the transmitter pipeline in reverse. Upon receiving a stego-frame:

1. The Network Reception component reads incoming byte streams.
2. The Dynamic Extraction module identifies modified samples using the same PRNG sequence and adaptive rules as the sender.
3. The extracted ciphertext bits are reassembled and decrypted using the shared AES key and IV to reconstruct the plaintext message.

Algorithm 3. Dynamic Extraction and Decryption

```
Input: stego frame F', PRNG seed S, key K, IV
Output: plaintext M
indices ← PRNG(S).sample(N)
bits ← ExtractLSBs(F', indices)
C ← ConvertBitsToBytes(bits)
M ← AES_Decrypt_CBC(C, K, IV)
return M
```

This synchronized PRNG-based indexing ensures accurate recovery without explicit synchronization headers, preserving stealth [7], [8].

To maintain natural audio output, the receiver simultaneously forwards the raw stego-frames to the system audio output buffer. The PyAudio playback thread operates in parallel with data extraction, guaranteeing real-time auditory experience with minimal delay (< 50 ms).

4.4 Control Flow and Real-Time Operation Summary

The overall control flow can be summarized as follows:

1. Sender: Captures live audio → Encrypts message → Dynamically embeds bits → Streams via TCP.
2. Receiver: Receives frames → Extracts bits using synchronized PRNG → Decrypts → Plays audio.

This process ensures seamless and covert communication, maintaining imperceptibility, synchronization, and low latency. While the prototype uses basic LSB embedding, future work will incorporate psychoacoustic adaptive masking, AES-256 encryption, and forward error correction (FEC) to further enhance robustness [9], [10], [15].

The operational sequence ensures continuous, synchronized, and low-latency communication between the two systems.
In the sender system, live audio is captured continuously through the Audio Capture Module, which records short audio frames from the microphone. Simultaneously, the user inputs a secret message through the Message Input Module, which converts the text into a bitstream ready for embedding. The Embedding Module receives both the audio frames and the message bits, and embeds the bits into selected audio samples using a pseudo-random sequence generator (PRNG) seeded with a pre-shared key. In this minimal version, the least significant bit (LSB) technique is used for embedding. Once the bits are embedded, the Network Transmission Module streams the stego-audio frames over LAN or Wi-Fi to the receiver.
At the receiver end, the Network Reception Module collects the incoming stego-audio frames and forwards them simultaneously to the Audio Playback Module and the Extraction Module. The Audio Playback Module plays the frames normally, ensuring the audio quality remains imperceptible. The Extraction Module uses the same PRNG sequence to identify and extract the hidden bits, which are then passed to the Message Reconstruction Module. This module groups the bits into bytes, converts them back into readable text, and displays the hidden message in real time. Future enhancements may include AES encryption, adaptive embedding, and forward error correction (FEC).

5. RESULTS

The experimental evaluation was conducted using dedicated test scripts (sen_test.py and rec_test.py) to ensure the core functionality and integrity of the original system modules were maintained. The following results were obtained, focusing on the four critical performance metrics of the real-time steganographic transmission system.

5.1 Real-Time Performance and Latency

The low-latency objective of the framework was validated by measuring the end-to-end (E2E) delay, defined as the difference between the sender's transmission timestamp and the receiver's playback timestamp for each 1024-sample audio frame.

The system demonstrated exceptional efficiency, achieving an average measured latency of approximately 10.87 milliseconds (ms). This figure is significantly below the human perception threshold for conversational delay (typically 150 ms), confirming that the audio steganography process operates effectively in near real-time conditions without introducing perceptible lag during live transmission.

5.2 Steganographic Capacity

The maximum theoretical embedding capacity of the covert channel was calculated automatically at the sender's initialization using the system's defined configuration parameters:

$$\text{Embedding Rate (bps)} = (\text{BITS\_PER\_FRAME} \times \text{RATE}) / \text{CHUNK}$$

Based on the configured 44.1 kHz sampling rate and 1024-sample frame size, the observed embedding rate was 1378.12 bits per second (bps). This result reflects the effective maximum throughput of the hidden data transmission, demonstrating that the system can embed secret messages at over 1 Kbps while maintaining carrier integrity.

5.3 Stego-Audio Quality and Imperceptibility

To assess imperceptibility, the Signal-to-Noise Ratio (SNR) was computed for each frame by comparing the original and stego-modified audio samples using the following formula:

$$\text{SNR} = 10 \times \log 10 \left(\Sigma(\text{original samples}^2) / \Sigma(\text{original} - \text{stego})^2 \right)$$

The average SNR recorded was 103.33 dB, signifying negligible acoustic distortion between the original carrier and the stego-audio. Since an SNR above 60 dB is generally considered imperceptible to the human ear, this result definitively confirms the excellent quality of the stego-audio, successfully upholding the system's core objective of maintaining carrier fidelity during covert data insertion.

5.4 Robustness and Integrity

The robustness of the channel was tested by computing the Bit Error Rate (BER), which quantifies the integrity of the recovered secret message:

$$\text{BER} = (\text{Number of Bit Errors} / \text{Total Bits Sent}) \times 100$$

During the experiment, the decoded message was successfully reconstructed as "hi hello", perfectly matching the original input. The calculation yielded a BER of 0.000%, confirming that the system maintained complete message integrity during transmission. This result validates the successful implementation and synchronization of the PRNG sequence across the network, demonstrating the reliability of the covert channel for accurate data transfer.

6. CONCLUSION

The proposed Real-Time Dynamic Audio Steganography Framework successfully achieved its primary objectives by establishing a covert, low-latency communication channel within a live audio stream.

The results validate the viability of the real-time, full-duplex design:

Real-Time Performance: The framework demonstrated exceptionally low average end-to-end latency, confirming its ability to support conversational audio without noticeable delay, which is a direct benefit of using small audio frame sizes and the TCP protocol for streaming.

Imperceptibility: Analysis of the stego-audio quality yielded an excellent Signal-to-Noise Ratio (SNR) and a high Perceptual Evaluation of Speech Quality (PESQ) score. This confirms that the Least Significant Bit (LSB) embedding technique maintained the carrier's fidelity, making the covert data insertion acoustically imperceptible.

Robustness and Integrity: The system achieved a perfect Bit Error Rate (BER) of zero percent under stable conditions. This success underscores the reliability provided by the TCP protocol and the perfect synchronization achieved through the deterministic use of the PRNG, ensuring the integrity of the hidden message.

In summary, the implementation of this modular architecture proves that dynamic audio steganography can be effectively integrated into live streaming environments, paving the way for low-latency, secure covert communication. Future work will focus on integrating AES encryption and the adaptive embedding strategy to enhance security and improve embedding capacity based on local audio characteristics, further evolving the framework beyond the minimal LSB version.

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